

4.2 Database Components (Tables, Records, Fields)

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔

CS XI

Chapter 4: Data and Analysis

Lecture 03 of 15

4.2 Database Components (Tables, Records, Fields)

LEARN IT

The basic structure is a **table** (also called a relation): rows and columns, like a spreadsheet, holding facts about one kind of thing (Students, Teachers, Books).

A **record** (row, tuple) is one complete instance — everything about Ali. A **field** (column, attribute) is one kind of fact — the whole `Name` column, or the whole `Grade` column.

Roll No	Name	Course	Grade
101	Ali	CS	A
102	Sara	IT	B+
103	Ahmed	CS	A

The row `101 | Ali | CS | A` is one **record**. The `Name` column is one **field**.

REMEMBER IT

Memory: **row = record = one person. column = field = one question on the admission form.**

COMMON MISTAKE

Do not mix two entities in one table (students and library books together). Each entity gets its own table; keys will link them later.

اردو میں سمجھیں

ٹیبل قطاروں اور کالموں میں ڈیٹا رکھتا ہے۔ ایک قطار ایک ریکارڈ (مثلاً علی کی پوری معلومات) ہے، ایک کالم ایک فیلڈ (مثلاً صرف نام) ہے۔

CHECK YOURSELF

1. How many records and how many fields are in the table above?
2. Is "CS" a record or a field value?

Answers: 1. 3 records, 4 fields. 2. A value stored in the Course field.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A table is:

- a. Rows and columns of one entity type
- b. A Python for-loop
- c. A cable
- d. A sprint

MCQ 2. A record is:

- a. One row
- b. One column
- c. The DBMS name
- d. A query

MCQ 3. A field/attribute is:

- a. One column
- b. The whole database
- c. A report
- d. A form

MCQ 4. A value in one cell is:

- a. A data item
- b. A relationship
- c. A schema
- d. A view

MCQ 5. Students table columns might be:

- a. id, name, class
- b. AND, OR, NOT
- c. HTTP, TCP
- d. Mean, π

MCQ 6. Many records share the same:

- a. Fields (structure)
- b. Primary key value
- c. OSI layer
- d. Random NAND

Extra short questions

1. Define table, record, field.
2. Give a 3-field student record example.
3. What is a data type of a field?
4. Why not put two entities in one messy table?

Extra long question

1. Design a Marks table: fields, types, one sample record, and which field will likely become the primary key.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. Table: entity set. Record: one instance. Field: one property.

2. 101 | Ali | 85

3. The kind of value: Number, Text, Date, Yes/No.

4. Duplication and update anomalies — later ER/normalisation.

Long

1. roll (Number, PK), name (Text), subject (Text), marks (Number). Sample 12, Ali, CS, 85.
Justify uniqueness of roll or roll+subject.

Next lecture: 4.3.1 Keys and Integrity Constraints